INCH-POUND

MS27662E
w/AMENDMENT 2
27 June 2025
SUPERSEDING
MS27662E
w/AMENDMENT 1
8 February 2008

DETAIL SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, THRU-BULKHEAD MOUNTING, BAYONET COUPLING, SERIES I

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

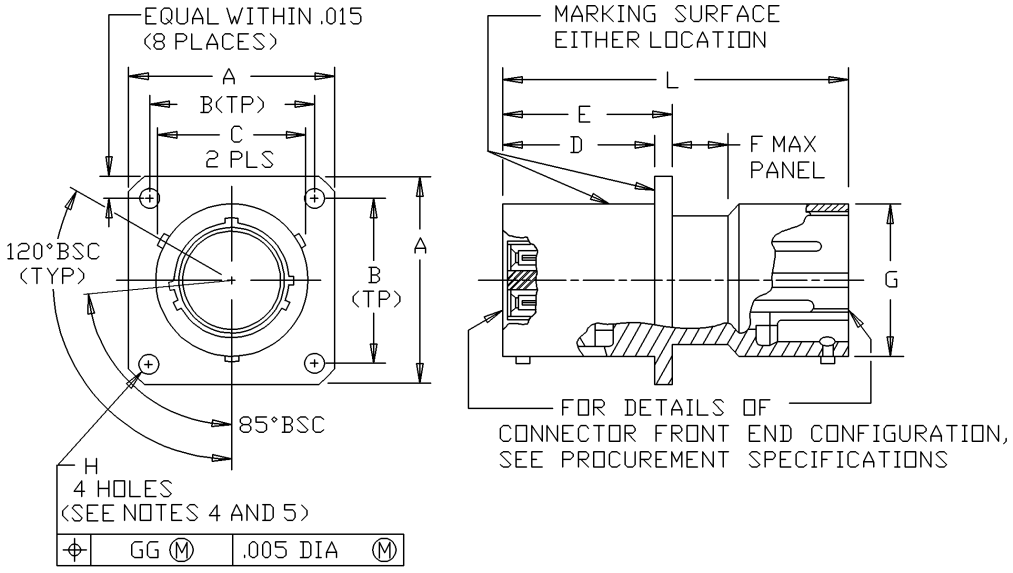


FIGURE 1. Receptacle, class E.

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Shell size	A ±.020	B (TP)	C +.000 -.016	D +.000 -.005	E +.006 -.001	F max 1/	G +.001 -.005	H +.010 -.005	L max
9	.938	.719	.662	.632	.717	.125	.572	.128	1.614
11	1.031	.812	.810				.700		
13	1.125	.906	.960				.850		
15	1.219	.969	1.085				.975		
17	1.312	1.062	1.210				1.100		
19	1.438	1.156	1.317				1.207		
21	1.562	1.250	1.442	.602			1.332		
23	1.688	1.375	1.567				1.457	.147	
25	1.812	1.500	1.692				1.582		

1/ F dimension excludes mounting hardware.

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.602	15.29	.938	23.83	1.207	30.66	1.457	37.01
.005	0.13	.632	16.05	.960	24.83	1.210	30.73	1.500	38.10
.006	0.15	.662	16.81	.969	24.61	1.219	30.96	1.559	39.60
.010	0.25	.700	17.78	.975	24.76	1.250	31.75	1.562	39.67
.011	0.28	.717	18.21	1.031	26.19	1.312	33.32	1.567	39.80
.016	0.41	.719	18.26	1.062	26.97	1.317	33.45	1.582	40.18
.125	3.18	.810	20.57	1.085	27.56	1.332	33.83	1.614	41.00
.128	3.25	.812	20.62	1.100	27.94	1.375	34.92	1.688	42.88
.147	3.73	.850	21.59	1.125	28.58	1.438	36.53	1.692	42.98
.572	14.53	.906	23.01	1.156	29.36	1.442	36.63	1.812	46.02

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. In example of part number, contact style C indicates feed through contacts.
4. The gauge features for GG shall be .010 (0.25 mm) smaller than the MMC size at basic location.
5. For main keyway and keyway positions, see MIL-DTL-38999.
6. Operating voltages are listed in table I for guidance.

FIGURE 1. Receptacle, class E - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Interface dimensions shall conform to MIL-DTL-38999.
Contact resistance shall be 125 percent of that specified in SAE-AS39029 .
Connectors shall show no leakage in excess of one atmospheric cubic inch of air per hour. (4.55×10^{-3} CM ^{3}/S) when subjected to a pressure differential of one atmosphere across the connector as specified in MIL-PRF-83178 for class T.

Operating voltage, see table I.

TABLE I. Operating voltage.

Service ratings	Operating voltage (sea level)	
	AC (rms)	DC
M	400	550
I	600	850
II	900	1250

Mating connectors: MS27467, MS27653 and MS27661.
Insert arrangements: See MIL-STD-1560.
Finish shall be in accordance with MIL-DTL-38999.

Part or Identifying Number (PIN) example:



no letter is required for normal.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MS27467
MS27653
MS27661
MIL-PRF-83178
MIL-STD-1560
SAE-AS39029

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2025-059)

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19

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